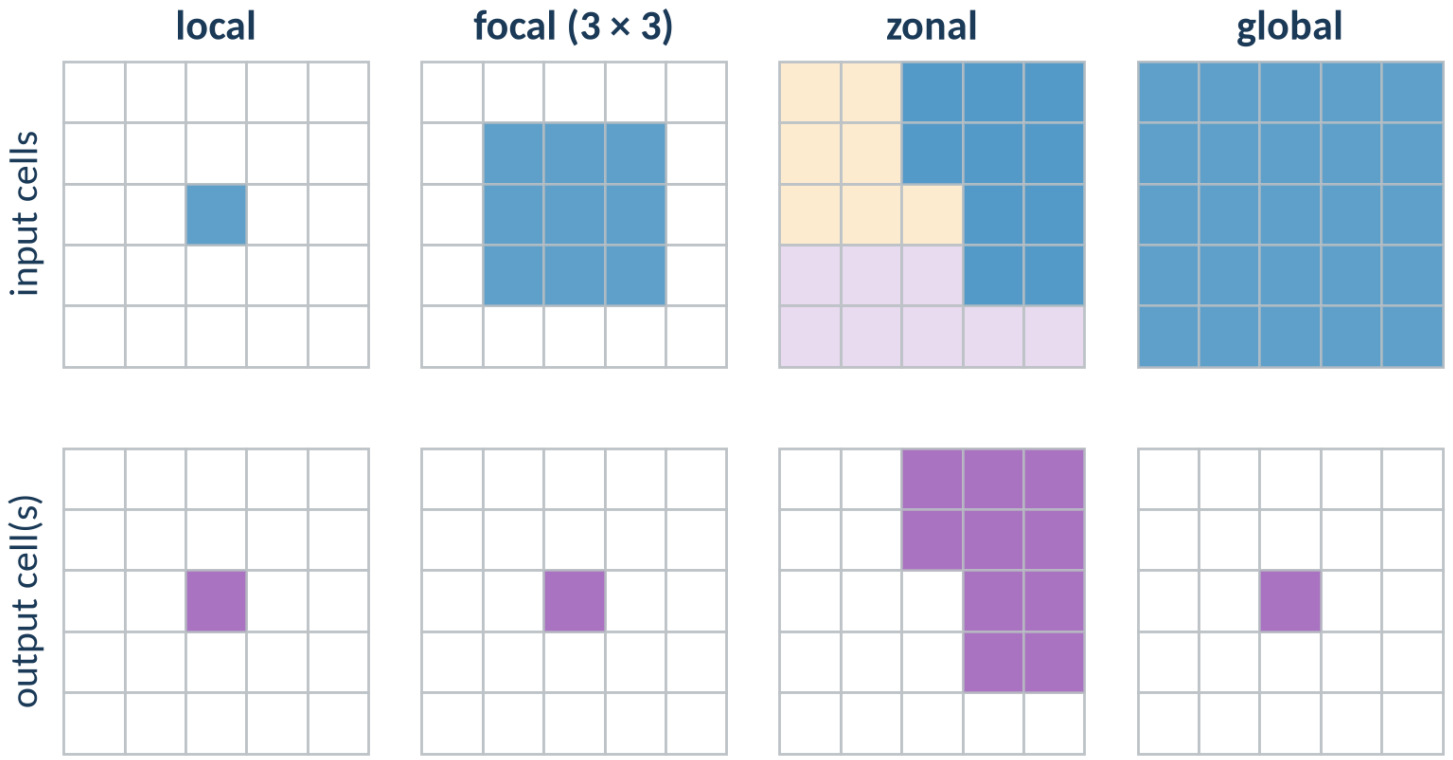
Raster operations

Map algebra: local, focal, zonal and global operations on aligned grids

Examples: Sentinel-2 and Copernicus DEM, Kamp-Lintfort



Map algebra: which cells feed the result?



- **Local:** cell by cell (NDVI, A + B)
- **Focal:** moving window (slope, smoothing)
- **Zonal:** per zone (mean NDVI per land cover)
- **Global:** whole grid (distance, min/max)

After C. D. Tomlin (1990): Geographic Information Systems and Cartographic Modeling

Local operation, cell by cell: NDVI

B04 red

185	179	205	228	257
181	218	218	247	460
179	234	336	681	2008
203	378	641	1286	2856
339	806	789	1036	2066

B08 NIR

3508	3352	4141	4446	4788
3960	4081	4389	4376	4232
4091	4540	4845	4152	3631
4371	4581	3507	2584	3266
4612	3428	1504	1399	2490

NDVI

0.90	0.90	0.91	0.90	0.90
0.91	0.90	0.91	0.89	0.80
0.92	0.90	0.87	0.72	0.29
0.91	0.85	0.69	0.34	0.07
0.86	0.62	0.31	0.15	0.09

orange cell: (4845 − 336) / (4845 + 336) = 0.87

In QGIS

Raster Calculator (band 4 = B08, 3 = B04):

("s2@4" - "s2@3") / ("s2@4" + "s2@3")

Contains modified Copernicus Sentinel-2 data 2025

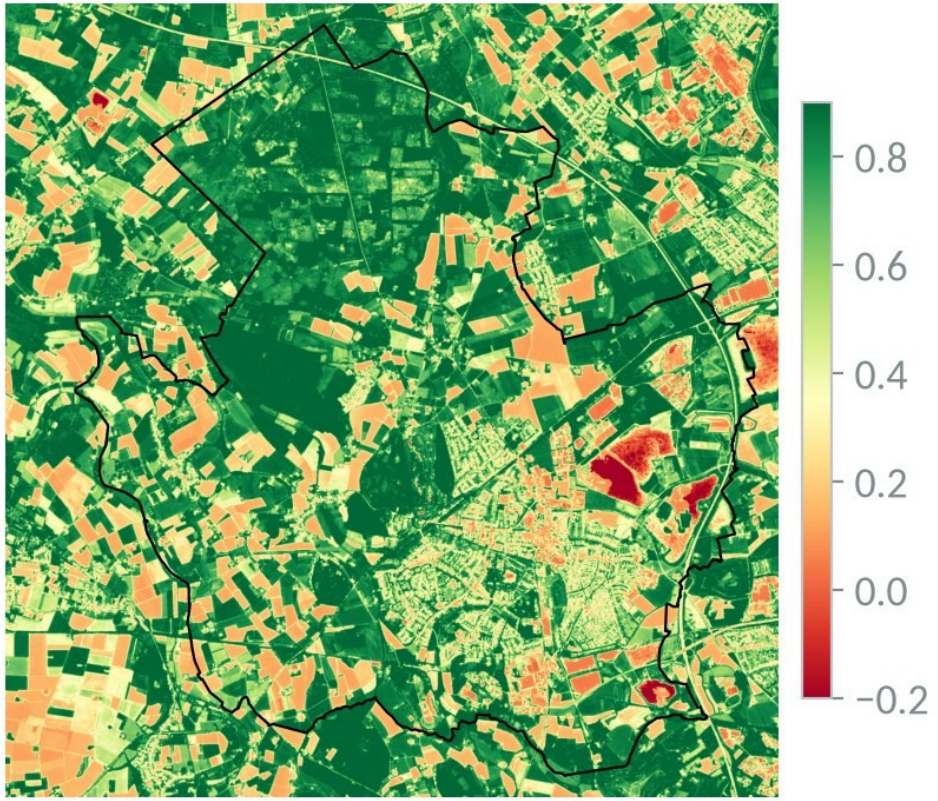


From a band stack to an index map

true colour (B04, B03, B02)



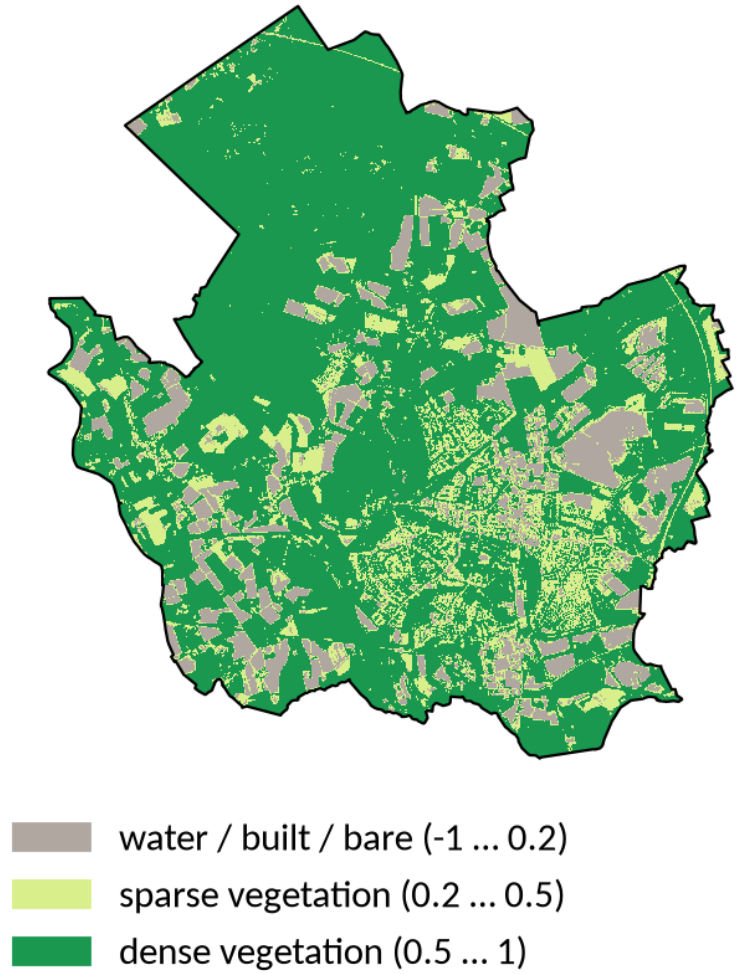
NDVI = (B08 − B04) / (B08 + B04)



In QGIS: Symbology → Multiband color: R = band 3, G = 2, B = 1

Contains modified Copernicus Sentinel-2 data 2025

Reclassify: conditions are local operations too



class	area [ha]
water / built / bare	853
sparse vegetation	1 027
dense vegetation	4 435

In QGIS

Raster Calculator:

```
("ndvi@1" >= 0.2)
  + ("ndvi@1" >= 0.5) + 1
```

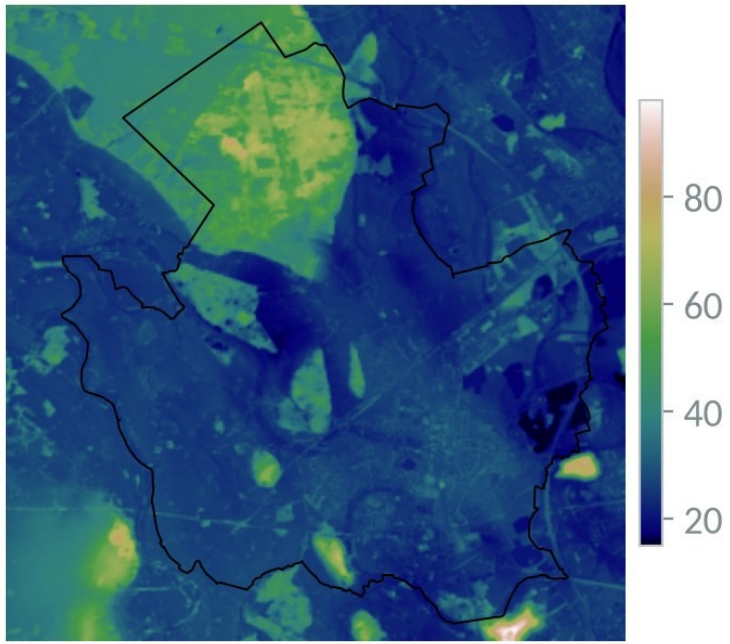
or Reclassify by table

Unique values report: cells

Contains modified Copernicus Sentinel-2 data 2025

Focal operations: the neighbourhood counts

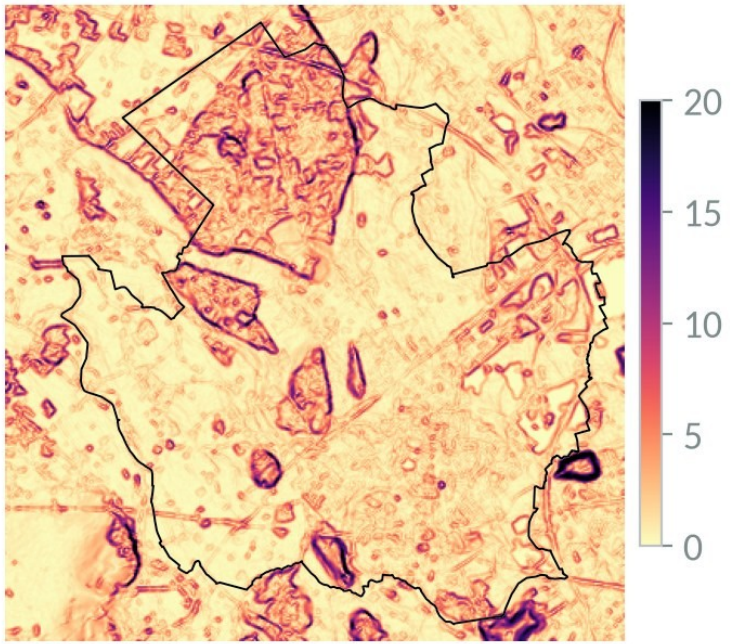
elevation [m] (DEM, 30 m)



hillshade (focal)



slope [°] (focal, 3 × 3)



In QGIS

Raster → Analysis → Slope, Hillshade

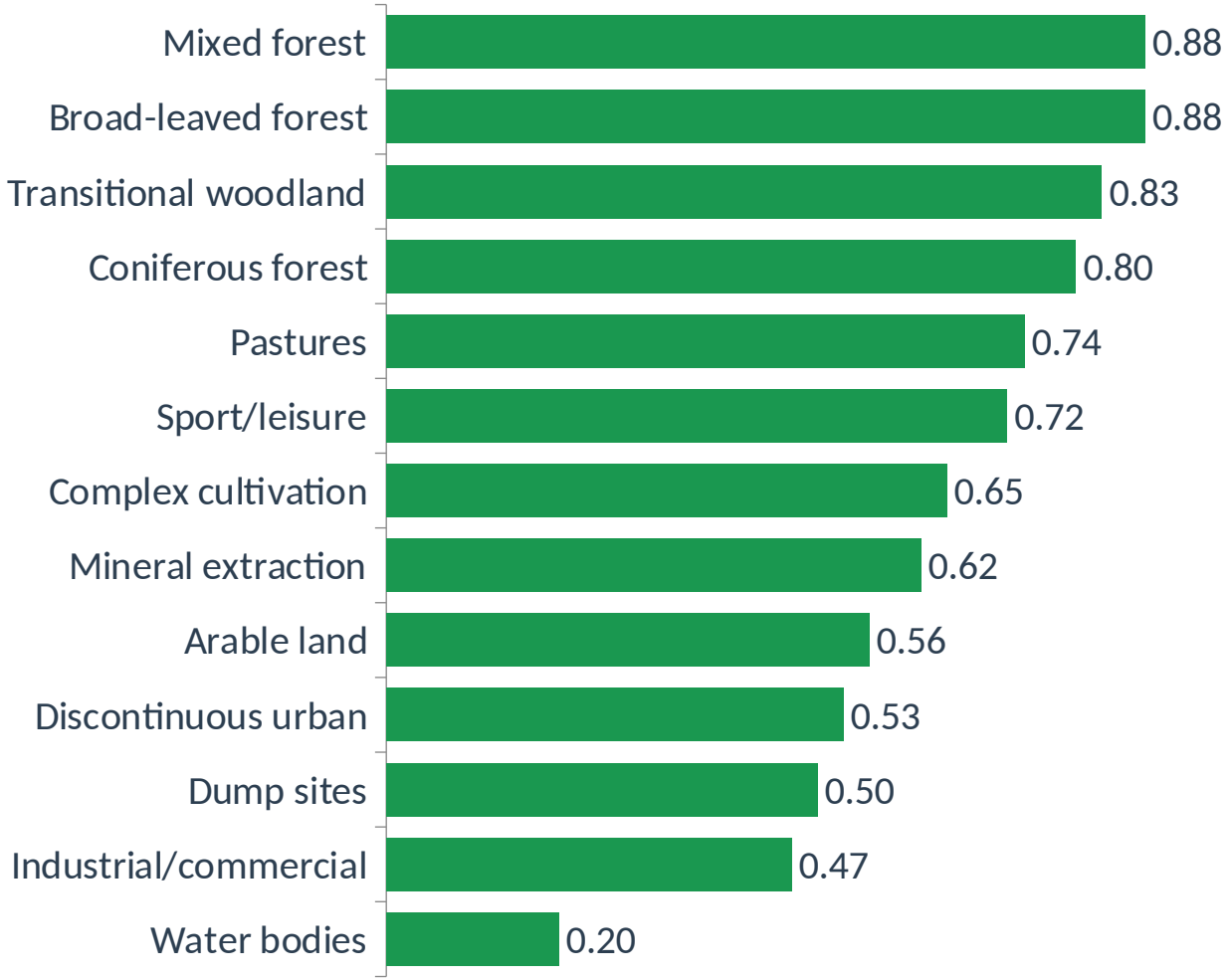
Processing → r.neighbors (3 × 3 mean)

3 × 3 mean kernel

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Copernicus DEM GLO-30, © DLR e.V. 2010-2014 and © Airbus 2014-2018, ESA

Zonal statistics: mean NDVI per land-cover zone



In QGIS

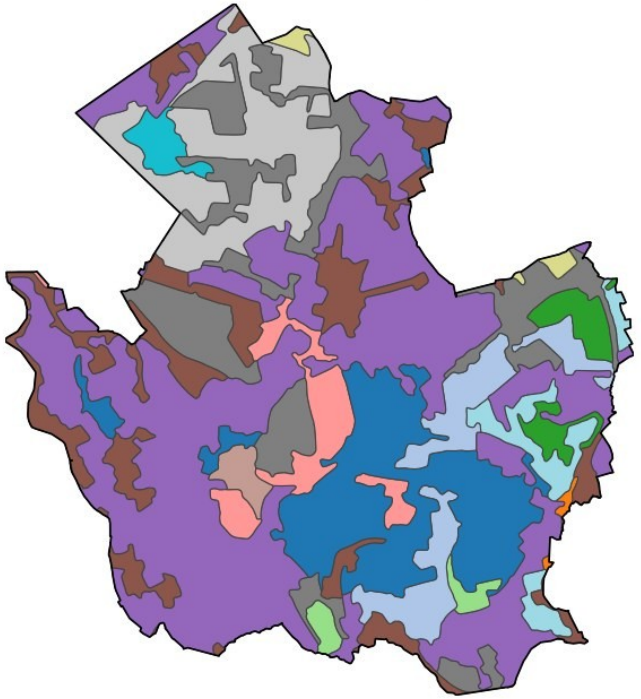
Processing →
Zonal statistics:
polygons = clc2018
raster = NDVI, mean
then Statistics by
categories (Code_18)

Vector zones are rasterised onto the value grid first.

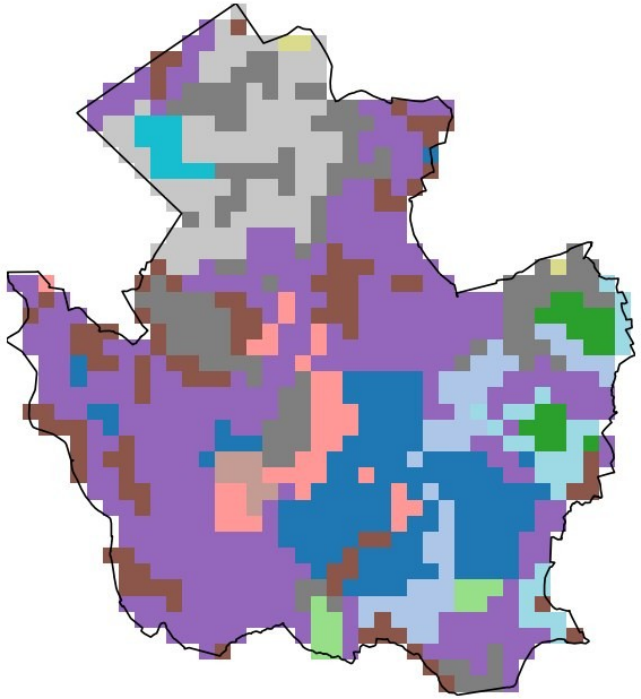
Contains modified Copernicus Sentinel-2 data 2025 · CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA

Converting between vector and raster

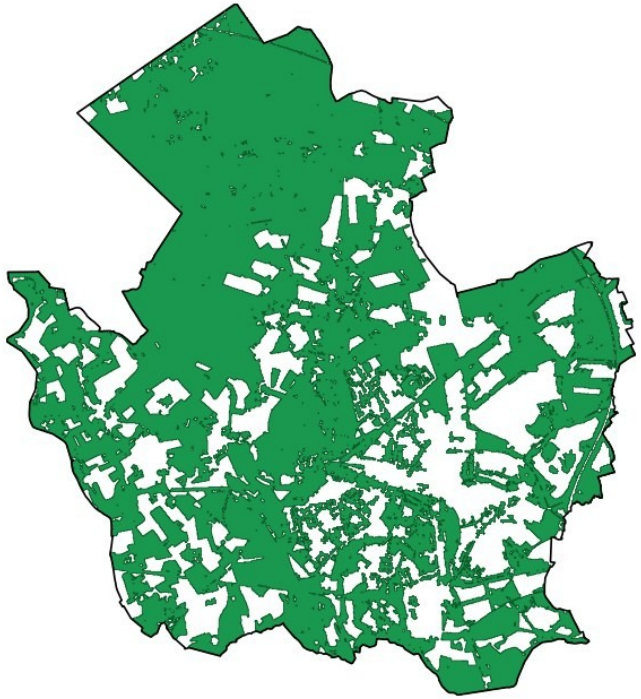
vector: 72 CORINE polygons



rasterize: 250 m cells



polygonize: 43 polygons ≥ 1 ha



In QGIS

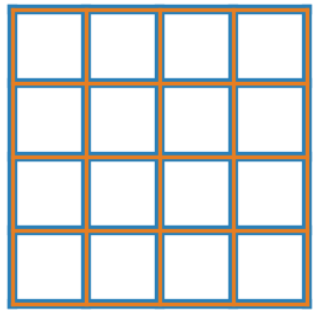
Raster → Conversion → Rasterize (vector to raster)

Raster → Conversion → Polygonize (raster to vector)

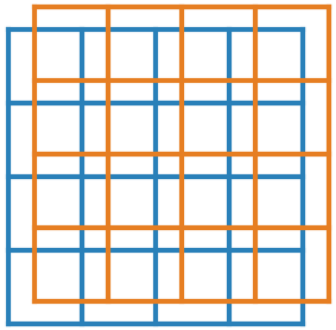
CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA · Contains modified Copernicus Sentinel-2 data 2025

Before combining rasters: align the grids

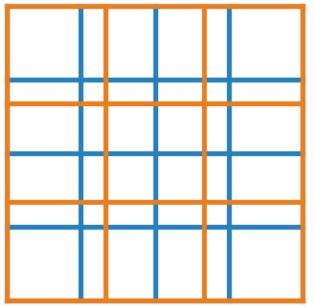
✓ aligned: same CRS, origin and cell size



✗ shifted origin



✗ different cell size



raster A raster B

In QGIS

Raster → Projections → Warp (Reproject)

Processing → Align rasters

- nearest: classes
- bilinear / cubic: continuous values
- never bilinear for land cover!

Raster operations in three tools

Operation	numpy / rasterio	QGIS	Type
band math, NDVI	(nir - red) / (nir + red)	Raster calculator	local
reclassify	np.select, np.where	Reclassify by table	local
smoothing, slope	scipy.ndimage, np.gradient	Slope, Hillshade	focal
zonal statistics	rasterize + groupby	Zonal statistics	zonal
distance to features	distance_transform_edt	Proximity	global
vector → raster	features.rasterize	Rasterize	conversion
raster → vector	features.shapes	Polygonize	conversion
align / reproject	WarpedVRT, warp.reproject	Warp, Align rasters	preparation



Sources and further reading

- **Data:** Contains modified Copernicus Sentinel-2 data 2025 (S2A_32ULC_20250513, Earth Search STAC); Copernicus DEM GLO-30, © DLR e.V. 2010-2014 and © Airbus 2014-2018, ESA; CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA; municipality boundary © OpenStreetMap contributors (ODbL)
- **Concepts:** C. D. Tomlin (1990): Geographic Information Systems and Cartographic Modeling, Prentice Hall
- **Further reading (links):** rasterio documentation; Earth Lab / ESIIL raster lessons (CC BY-SA); PyGIS raster chapters; QGIS Training Manual – Raster analysis
- **Figures:** own work, generated by make_slides.py in this unit

